EELU
الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

AI Powered Biomedical Equipment Management System

Graduation Project 2025 / 2026

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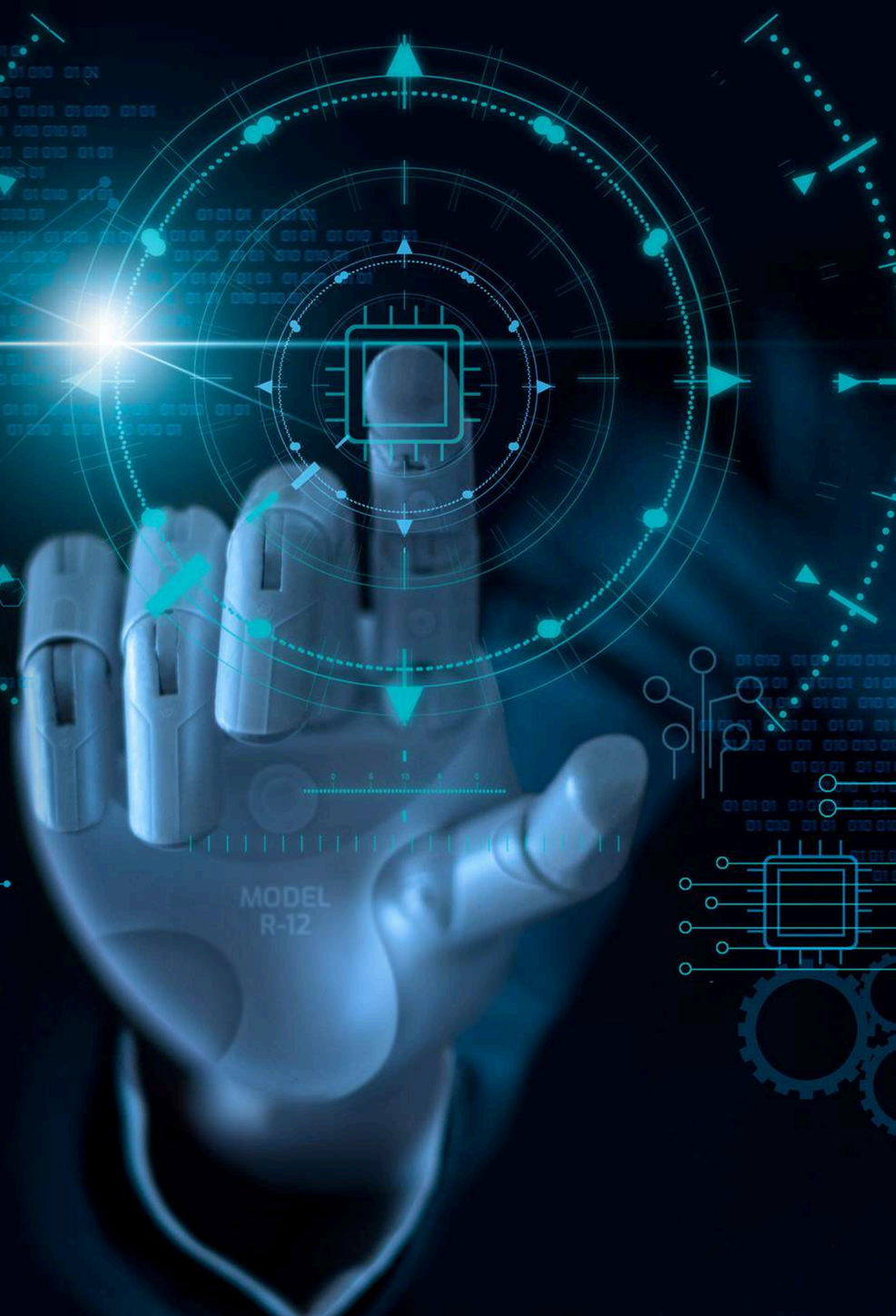

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1 Introduction & Motivation

2 Problem Statement

3 Related Work

4 Data Overview

5 System Overview

6 Methodologies

7 Demo

8 Conclusion

9 Future Work

10 References

Introduction

01 Healthcare Facilities
02 Quality Accreditation & standards
03 Motivation

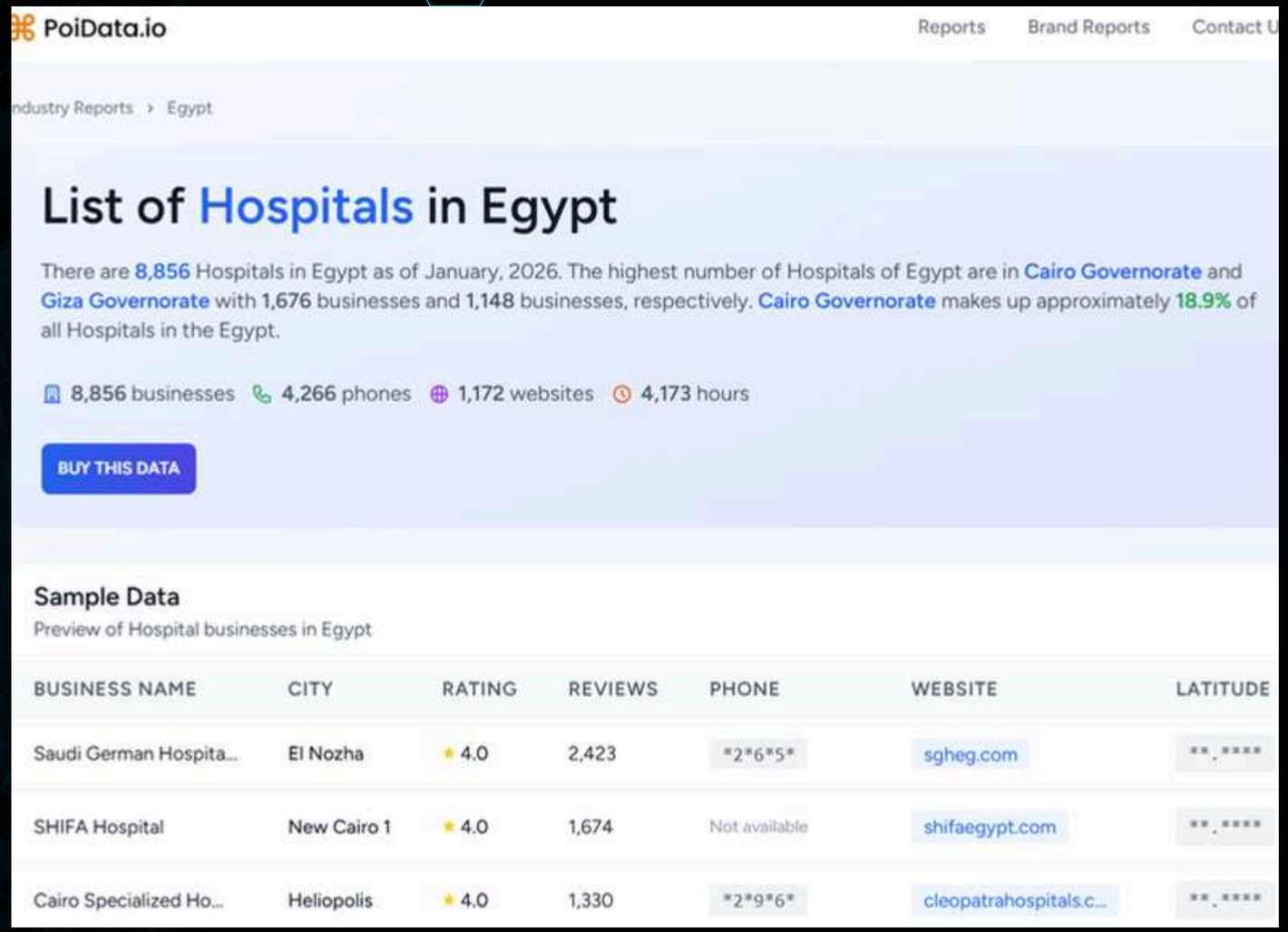
Healthcare Facilities

- **Egypt has around 8,856 hospitals as of 2025.**

reported by a data aggregator site, which compiles business listings (including private hospitals, clinics, and other healthcare facilities) : poidata.io

- Examples:

1. Saudi German Hospital ★4.0
2. SHIFA Hospital ★4.0
3. Ain Shams University ★3.0
4. Mabaret Al-Asafra ★3.0
5. Smouha Center ★3.0



Accreditation in Egypt

- **Around 196 healthcare facilities achieved full GAHAR accreditation.**

  The Egypt Healthcare Authority (EHA) announced that, out of 678 healthcare facilities assessed by GAHAR, 196 received full accreditation under nationally and internationally recognized quality and safety standards on 26 July 2025.

- This means only ~3% of health facilities have formal accreditation under recognized quality standards.

Accreditation in Egypt

Why Quality Accreditation Matters

Benefits for Patients

- Accreditation ensures facilities follow international safety and clinical quality standards, reducing errors and enhancing overall care quality

Benefits for Hospitals

- Accreditation shows that the hospital follows high-quality standards, which makes patients trust it more and improves its image and reputation in the community

Accreditation in Egypt

Why It Is Limited

Reasons Many Healthcare Facilities Lack Accreditation

- Lack of established systems for infection control, medical device management, and quality documentation
- Limited resources and trained staff to implement strict quality requirements
- Rapidly evolving standards: National guidelines like GAHAR continuously update to match international best practices, making compliance challenging for some facilities.

Motivation

Enhance hospital safety and reliability.

- Many hospitals still use manual or disconnected systems, causing delays, errors, and safety risks.

- Poor equipment monitoring leads to higher maintenance costs and potential patient harm.

Motivation

Enhance hospital safety and reliability.

- Our motivation is to bridge this gap by developing a smart, data-driven system that improves equipment management, ensures timely maintenance, and supports compliance with quality standards.
- This enhances patient safety, hospital reliability, and public trust

Problem Statemnet

01 Medical Equipment Challenges

02 Infection Control Challenges

Medical Equipment Challenges

- Many hospitals struggle to track equipment usage, status, and availability in real time.
- Maintenance is often reactive instead of predictive, leading to sudden failures.
- Lack of organized digital data makes it difficult to plan repairs and replacements.
- Manual records increase the risk of errors and missing information.
- Limited access to high-quality datasets reduces the effectiveness of AI solutions.

Infection Control Challenges

Infection control lacked consistency and proper tracking.

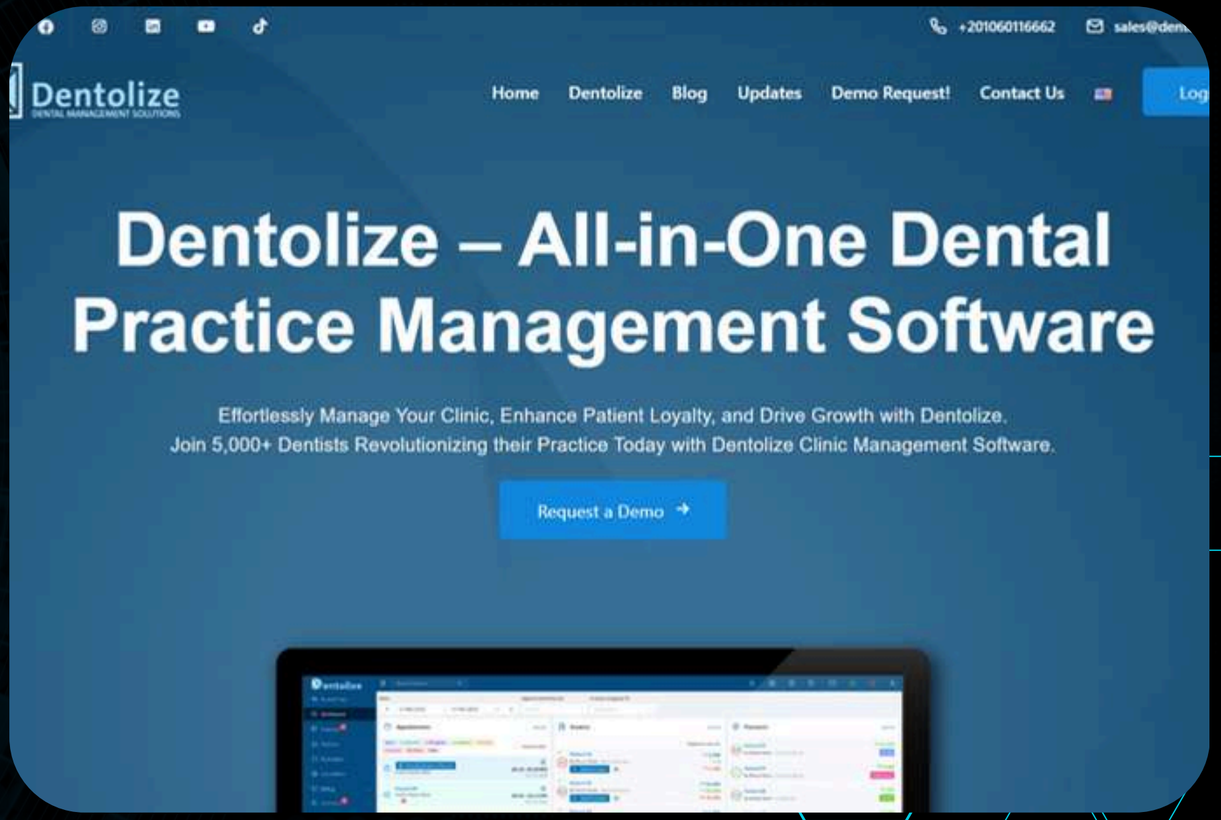
- Inconsistent monitoring of cleaning and sterilization procedures.
- Weak documentation of schedules and compliance.
- Limited digital tracking of infection-related indicators.
- Difficulty in identifying risk areas and preventing outbreaks early.

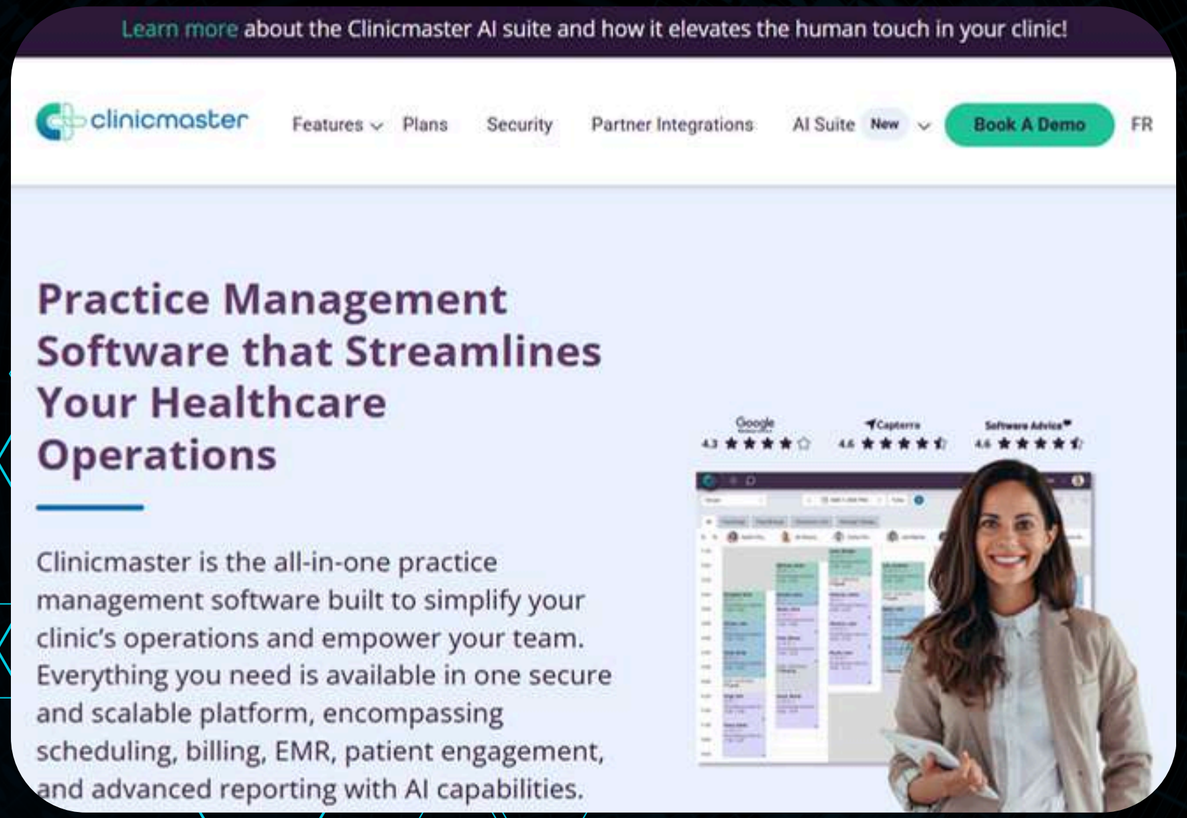
Related Work

Dentolize Clinic Management System

A system used to manage dental clinics, including patients, appointments, and billing

- Efficient management of patient records, appointments, billing, and basic clinic operations
- Lacks AI features, predictive maintenance, and advanced equipment or inventory management





Clinic Master

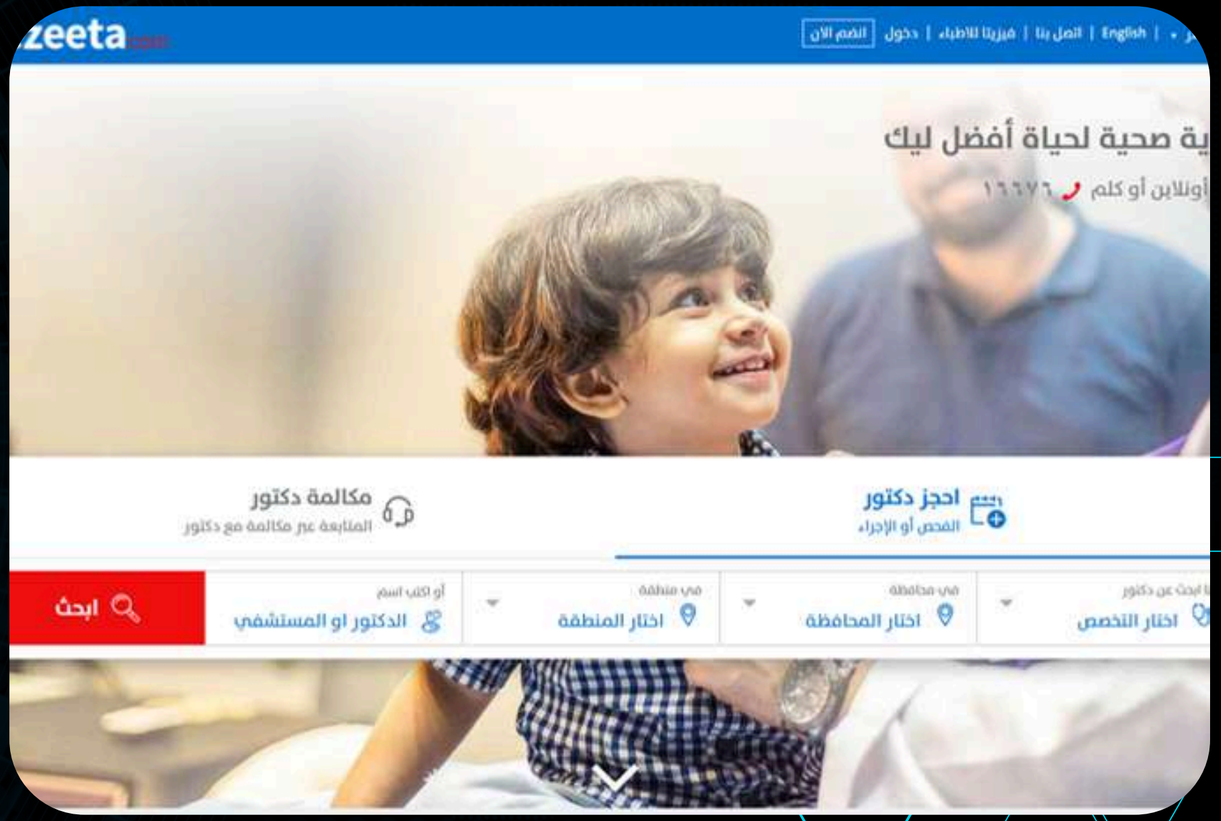
A healthcare management system for small to medium medical centers, handling scheduling, patient data, and basic reporting.

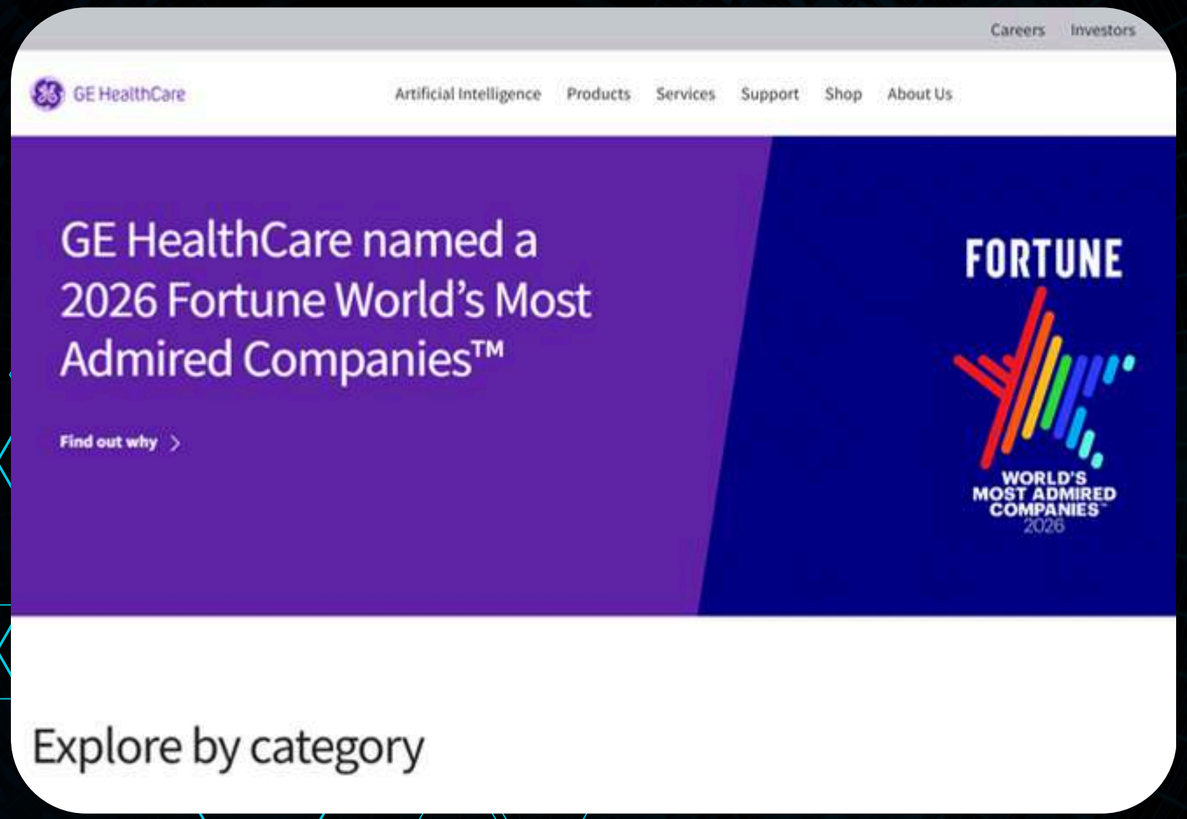
- ✅ Supports daily clinic operations including scheduling, patient records, and basic reports
- ❌ Limited equipment tracking with no usage analytics or preventive maintenance features

Vezeeta

A healthcare booking and patient engagement platform connecting patients with doctors.

- Improves appointment booking and patient engagement efficiently
- Does not support clinic operations such as equipment inventory, maintenance, or patient safety management





(International System)

GE Healthcare

An AI-driven healthcare analytics platform providing predictive insights and operational optimization.

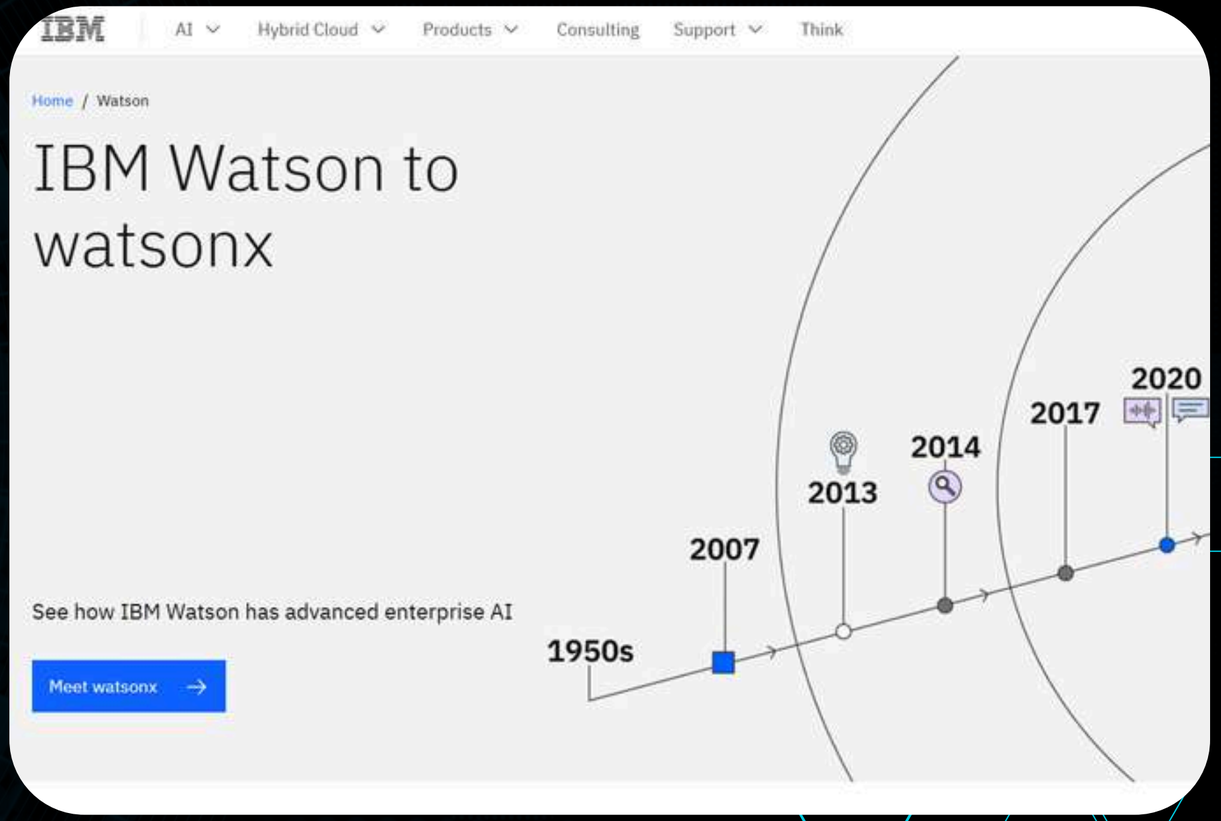
- Advanced AI analytics for prediction and operational efficiency
- High cost and complex infrastructure make it impractical for most Egyptian hospitals

(International System)

IBM Watson Health

An enterprise healthcare system for asset and medical equipment management integrated with devices.

- Powerful enterprise-level equipment and asset management
- Requires proprietary integrations and expensive service contracts, limiting use in cost-sensitive markets



Common Disadvantages Across Systems

01 No AI-based predictive maintenance causing Reactive maintenance instead of preventive maintenance

02 Inventory shortages detected too late

03 Poor visibility of equipment lifecycle and aging

04 High costs limiting adoption in Egypt

05 Complex infrastructure requirements

How Our Project Fills the Gaps?

- Predicts equipment failure using historical usage and maintenance data & Prioritizes maintenance tasks based on risk scores
- Optimizes inventory levels to prevent shortages
- Shows equipment condition indicators
- Designed to be cost-effective and scalable for Egyptian hospitals
- Does not require proprietary hardware or expensive licenses
- Alignment with International & National Standards like (ISO & GAHAR)

Value Proposition (Optional):

A locally-adapted, cost-efficient AI system that transforms medical equipment management from reactive to predictive while enhancing patient safety in Egyptian healthcare facilities.

Data Overview

01 Data Challenges

02 Data Preparation & Challenge Resolution

03 Final Dataset Structure

Data Challenges

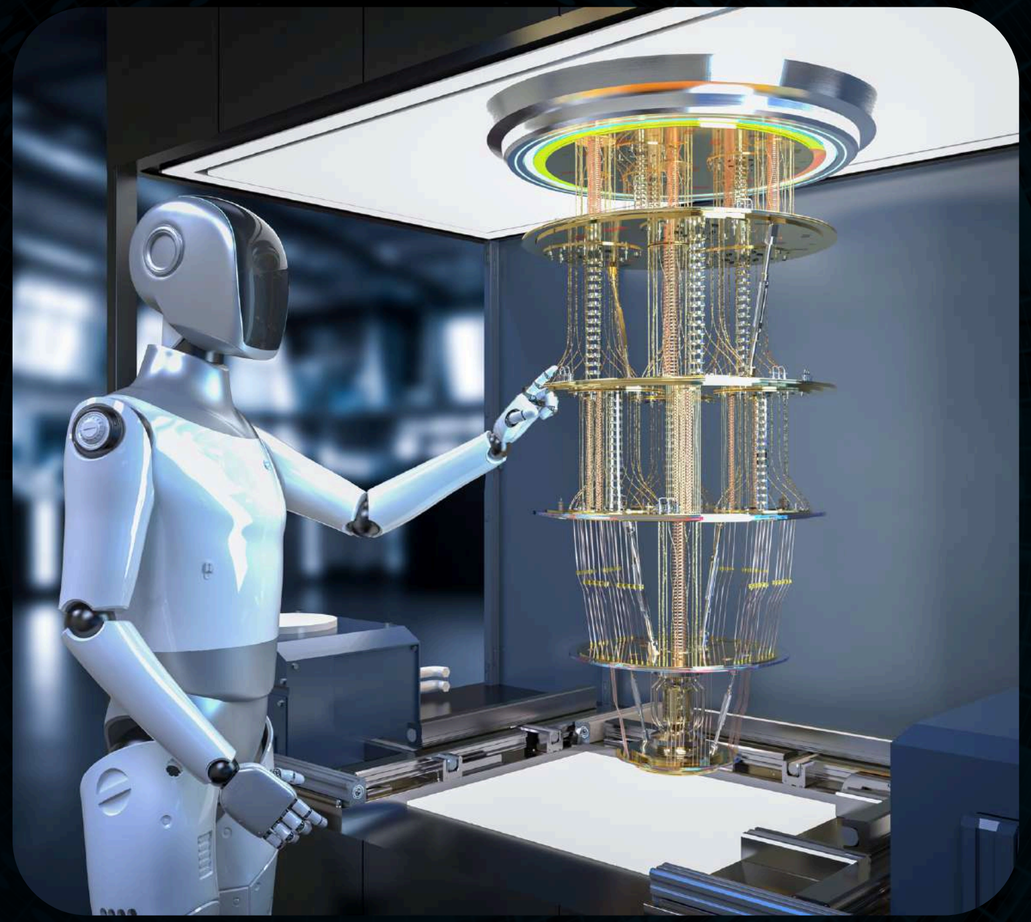
Limited data availability.

01 Depending only on online data was not sufficient for our project.

02 Publicly available datasets in this medical field are very limited and incomplete.

03 Access to high-quality medical data is restricted due to privacy regulations.

04 Some data was stored manually or in outdated systems.



Data Preparation & Challenge Resolution

- To overcome data rarity, we adopted a field-based data acquisition approach.
- We visited multiple hospitals and medical centers (Smouha Medical Center).

Data Preparation & Challenge Resolution

We held technical and administrative meetings with:

- 01 Biomedical Engineers
- 02 Hospital IT Manager

After negotiations and approvals, we obtained authorized datasets from:

- 01 Mabaret Al-Asafra Hospital
- 02 Smouha Medical Center

We conducted extensive online research and collected available resources, including:

- 01 Medical equipment reference books
- 02 GAHAR standards and guidelines
- 03 Technical documentation

Final Dataset Structure

Medical equipment dataset.

- The dataset contains detailed records of medical equipment collected from hospitals.
- Fields cover device identity, technical specifications, purchase & warranty info, maintenance, and performance metrics.
- Data is shown as :

01 Device Identity

02 Device tech info

03 Performance & Maintenance Metrics

Device Identity

DEVICE NAME

The name of the equipment is crucial for identification and reference in documentation.

BRAND

The manufacturer or brand represents quality assurance and reliability in device performance.

MODEL

The model name or number specifies the version and features of the device.

SERIAL NUMBER

A unique identifier assigned to each device for tracking and support purposes.

CODE

An internal device code used for inventory management and logistical processes.

CRITICAL / NON-CRITICAL

This classification indicates the device's importance in hospital operations and patient care.

Device Identity

Critical/Non-critical device	Code	Serial number	Model	Brand	Device name
Not embarrassing	035-KAW-03	011330 0021	PICCOLIGHT	PICCOLIGHT	OTOSCOPE
critical	051-AEO-01	SL510EY4ZR014	Shangr ila 510	AEONMED	Artificial respiration
critical	051-WEI-01	20,695	MEDUMAT Standard a	WELNMANN	Artificial respiration
Not embarrassing	015-MIN-01	FK-93017400	BeneHeart R3	MINDRAY	drawing a heart
Not embarrassing	015-MIN-09	FK-94018250	BeneHeart R3A	MINDRAY	drawing a heart
Not embarrassing	015-SHE-01	103B13808167E12	ECG-1103	CareWell	drawing a heart
critical	027-BLT-01	B073	P500	BIOLIGHT	syringe solutions

Device Info

PURCHASE DATE

The date when the device was officially purchased and registered for use.

YEAR OF PURCHASE

Indicates the year the device was acquired for calculating its age and depreciation.

DEVICE LIFE

The expected lifespan of the device, along with its actual operational life recorded.

PURCHASING POWER

Refers to the budget allocated for the procurement of this specific device.

WARRANTY PERIOD

The coverage timeframe for repairs and services, including start and end dates.

WARRANTY CONTRACT

Details regarding the warranty and maintenance agreements established with the supplier.

Device Info

End of warranty	Warranty start	Warranty period	purchasing power	Supplier company	Device life	Current year	Year of purchase	hnical competence %	Purchase date
				Fatimid		2024			
				Al-Athi	8	2024	2016		2016
27/11/2022	28/11/2021	General warranty		Sergey Tek	3	2024	2021		2021
				TechnoWave	5	2024	2019		2019
				TechnoWave	5	2024	2019		2019
				Intermed Hospital		2024			
				Capital	6	2024	2018		2018
				Ghtouri	8	2024	2016		2016
				Al-Athi	8	2024	2016		2016
27/11/2022	28/11/2021	General warranty		Sergey Tek	3	2024	2021		2021

Performance & Maintenance Metrics

DEVICE FAILURE RATE

The average device failure rate indicates reliability and potential operational interruptions.

TECHNICAL RATING

A technical rating assesses the device's overall performance and quality standards.

SPARE PARTS AVAILABILITY

Spare parts availability ensures timely repairs and minimizes downtime for hospital operations.

FUNCTIONALITY

Functionality measures how effectively the device performs its intended tasks in practice.

JOB PERFORMANCE

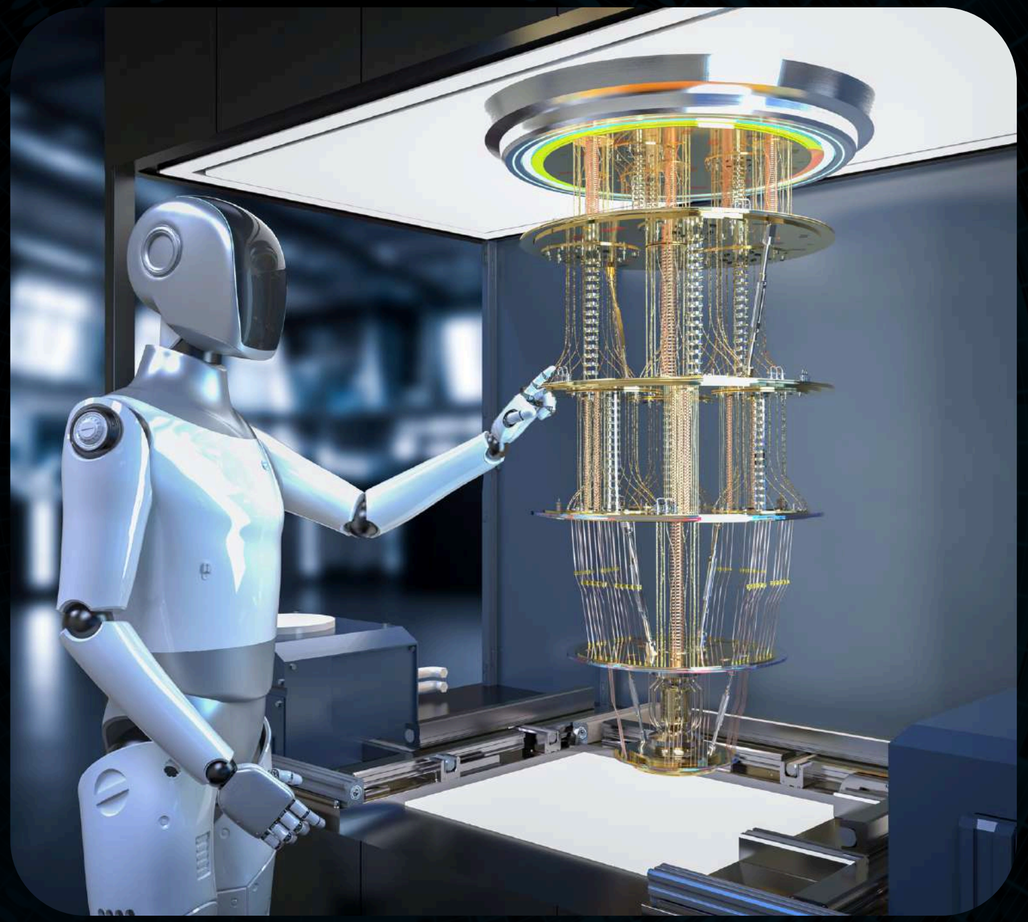
Job performance reflects the efficiency of the device in its operational environment.

APPARENT STATE

The apparent state evaluates the visual and functional condition of the device over time.

Performance & Maintenance Metrics

Functionality 100	are parts availability 20	Technical rating 30		erage device failure rate: 30%	operational life
		Job performance 20	Apparent state 10		
75	20	20	10	25	
87	20	20	10	20	17
100	20	20	10	30	20
100	20	20	10	30	20
100	20	20	10	30	20
73	20	20	8	25	
99	20	20	10	30	19



Wrapping Up

- The current dataset is not perfect for the training
- The dataset will require significant refinement and tuning before training.
- Our focus in the next phase will be on optimizing data quality and preprocessing.
- Goal: Achieve the best possible model performance through careful training and fine-tuning.

System Overview

01 Expected Features (Functional Requirements)

02 Non-Functional Requirements

Expected Features (Functional Requirements)

01 Core Inventory & Asset Management

- Digital Asset Records for all equipment with full metadata and manuals
- Real-time location and status tracking (In Use, Maintenance, Decommissioned).

02 AI-Driven Predictive Maintenance

- Predict failures 72+ hours in advance
- Smart scheduling based on usage and device health
- Anomaly detection for faults or unusual behavior.

Expected Features (Functional Requirements)

03 Utilization Analytics & Procurement

- See which equipment is used a lot or a little(Heat Maps).
- Predict when new equipment or spare parts needs to be bought.

04 Compliance & Risk Management

- Highlight high-risk devices for inspection first.
- Automatic compliance report generation(ISO, GAHAR).
- Prevent use of unsafe devices (mark equipment as unavailable if it has critical defects)

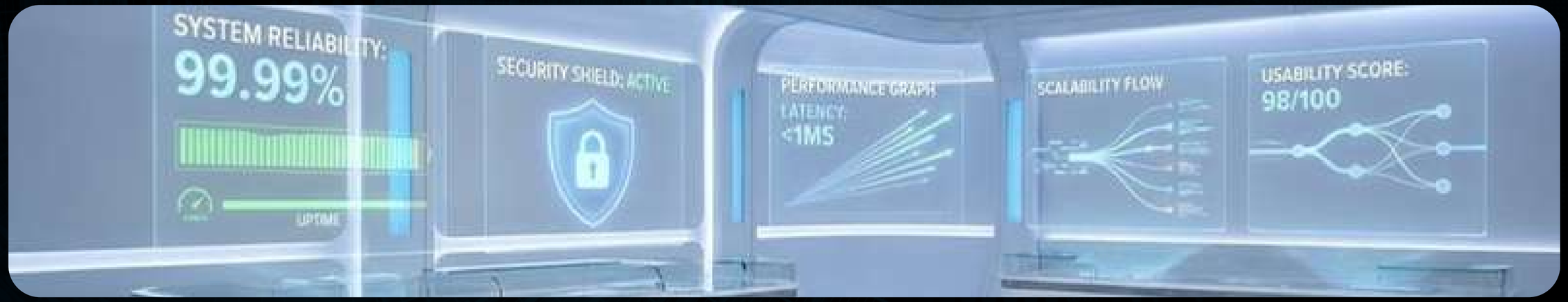
Non-Functional Requirements

01 Security & Privacy

- Keep data safe with encryption and secure access
- Track all actions with audit logs.

02 Performance & Reliability

- Always available, fast alerts for failures.
- Can handle lots of equipment without slowing down.



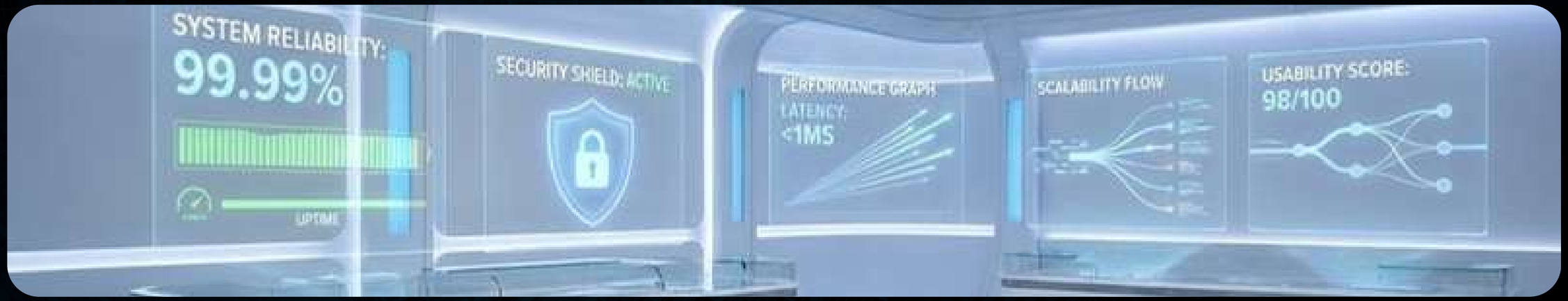
Non-Functional Requirements

03 Usability

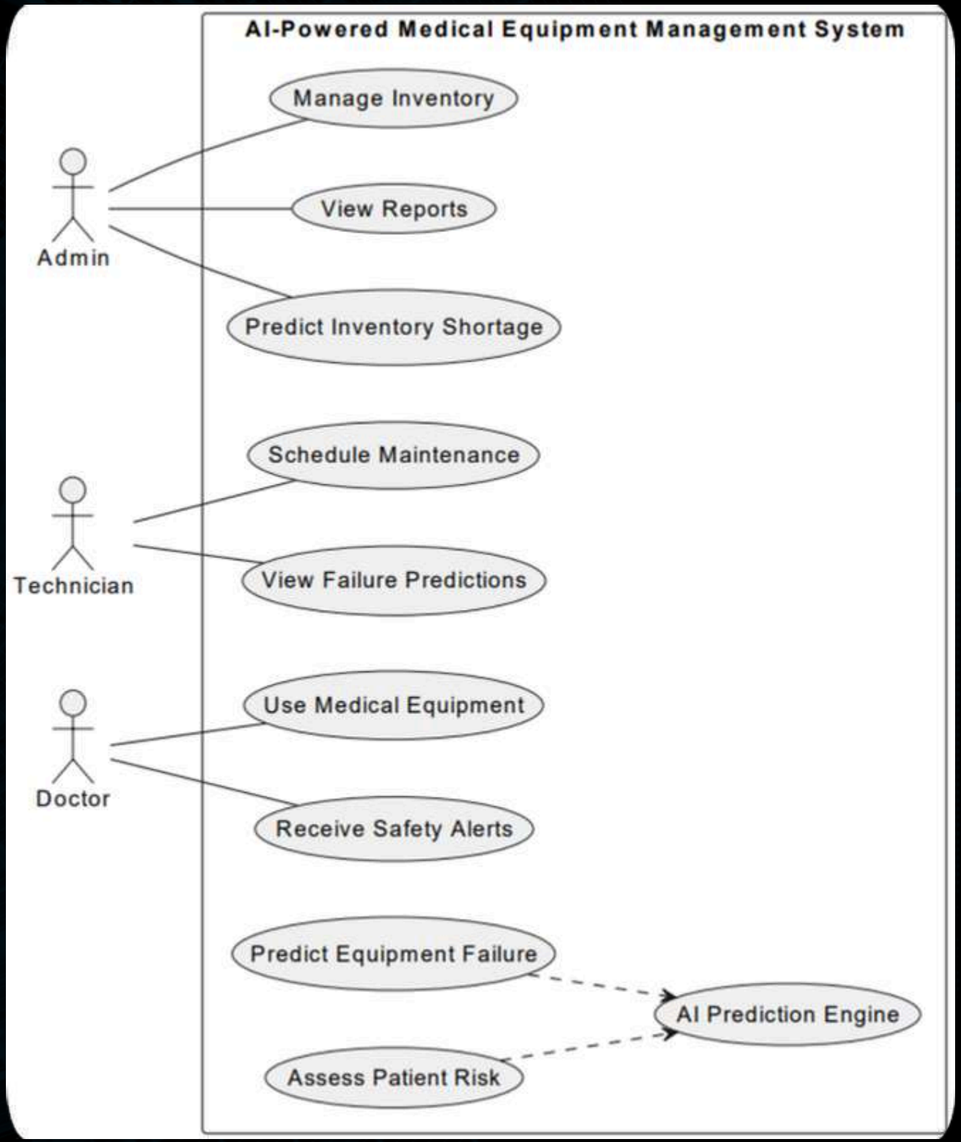
- Easy for staff to use and reduces workload.
- Can record data offline and sync automatically.

04 Interoperability

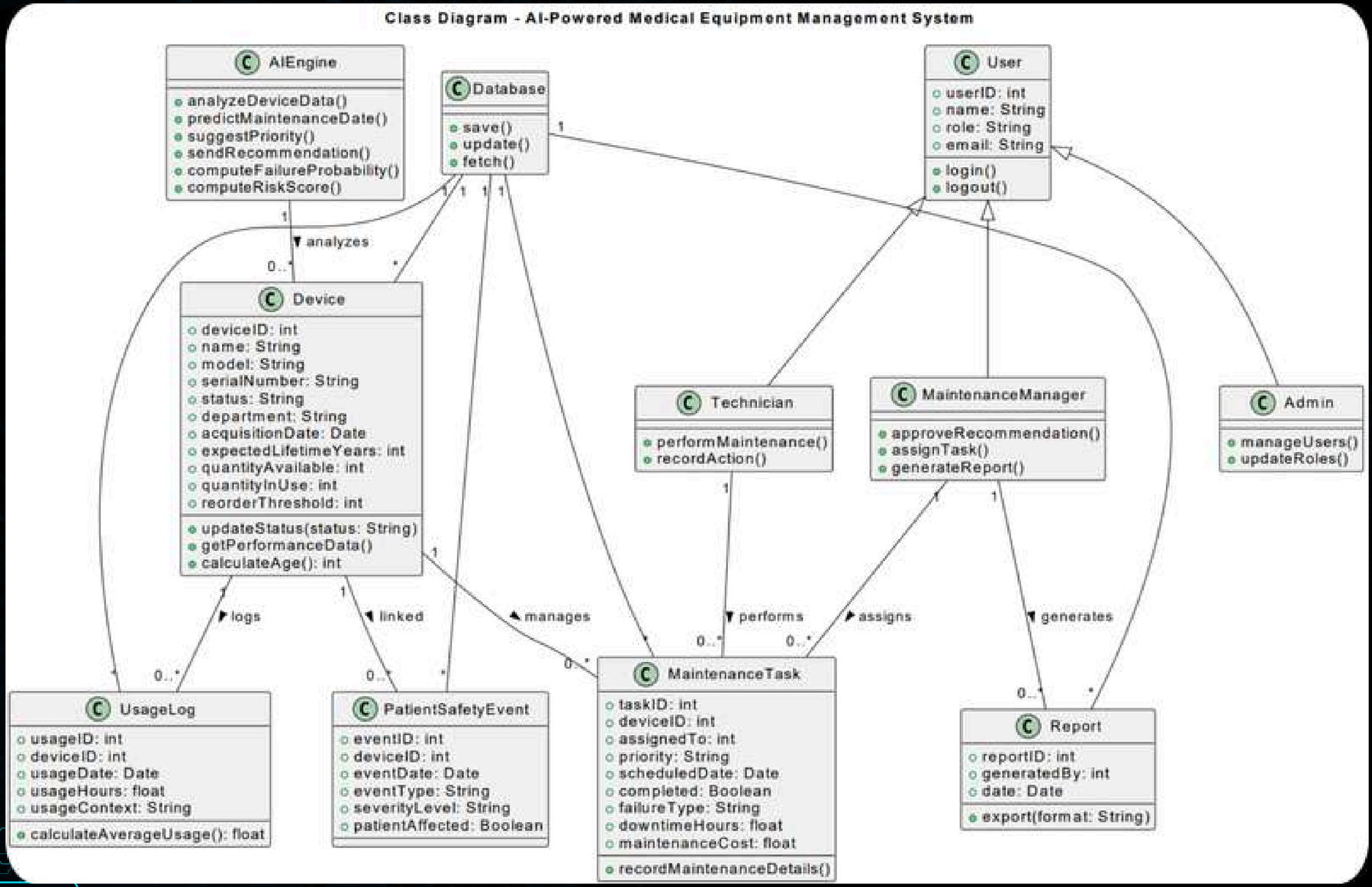
- Can work with hospital systems (EHRs)
- Connects easily to IoT devices using standard protocols.



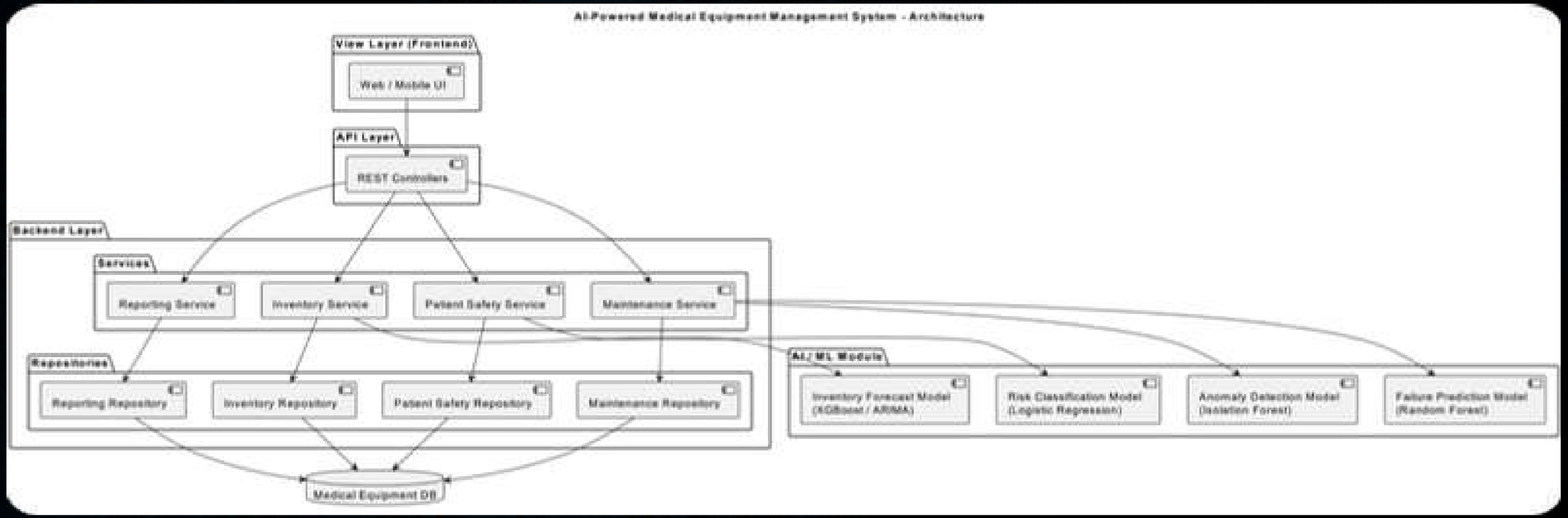
Use Case Diagram



Class Diagram



System Architecture Diagram



Methodologies

Methodologies

UI / UX

Why Figma?

- Cloud-based collaboration
- Real-time team updates
- Fast prototyping & wireframing
- Easy design handoff to React developers

Frontend

Why React.js?

- Component-Based Architecture
- High Performance
- Scalability
- Strong Integration
- Large Ecosystem

Methodologies

Backend

Why Java - Spring Boot?

- Enterprise-grade backend framework
- High reliability for critical systems
- Rapid development with built-in configurations
- Easy REST API creation
- Seamless integration

AI

Why Python?

- Best language for AI & data analysis
- Powerful ML libraries (Scikit-learn, TensorFlow)
- Efficient data processing
- Easy integration with Java backend via REST

Methodologies

Security

Why Spring Security?

- Industry-standard security framework
- Designed for enterprise and healthcare systems
- Deep integration with Spring Boot
- Highly configurable authentication and authorization mechanisms

Database

Why MySQL / PostgreSQL?

- Industry-standard relational database
- ACID compliance for strict data integrity
- Seamless integration with Spring Data JPA
- High performance and scalability

Methodologies

- UI/UX Design
- Frontend Development
- Backend Development
- Database Development
- AI & Data Processing
- Security Development

Techniques

1. Centralized Asset Management System

- **Tools**: PostgreSQL / MySQL,
- Stores equipment metadata, manuals, and lifecycle records while enabling fast inventory control and real-time status tracking.

2. Smart Device Identification & Tracking

- **Tools**: QR / RFID, OpenCV, REST APIs
- Provides instant device identification and location monitoring, reducing loss and improving availability

3. AI-Based Predictive Maintenance

- **Frameworks**: TensorFlow, PyTorch, Scikit-learn
- **Models**: LSTM, Random Forest, XGBoost
- Predicts failures 72+ hours in advance and detects abnormal performance patterns for smart scheduling

4. Utilization Analytics & Heat Mapping

- **Tools**: Pandas, NumPy, Power BI / Tableau, Matplotlib
- Analyzes historical usage to generate heat maps and support procurement planning.

Techniques

5. Automated Compliance & Risk Analysis

- **Tools**: Rule Engines, Python Scripts
- **Standards**: ISO 13485, GAHAR
- Automatically generates compliance reports and prioritizes high-risk devices for inspection.

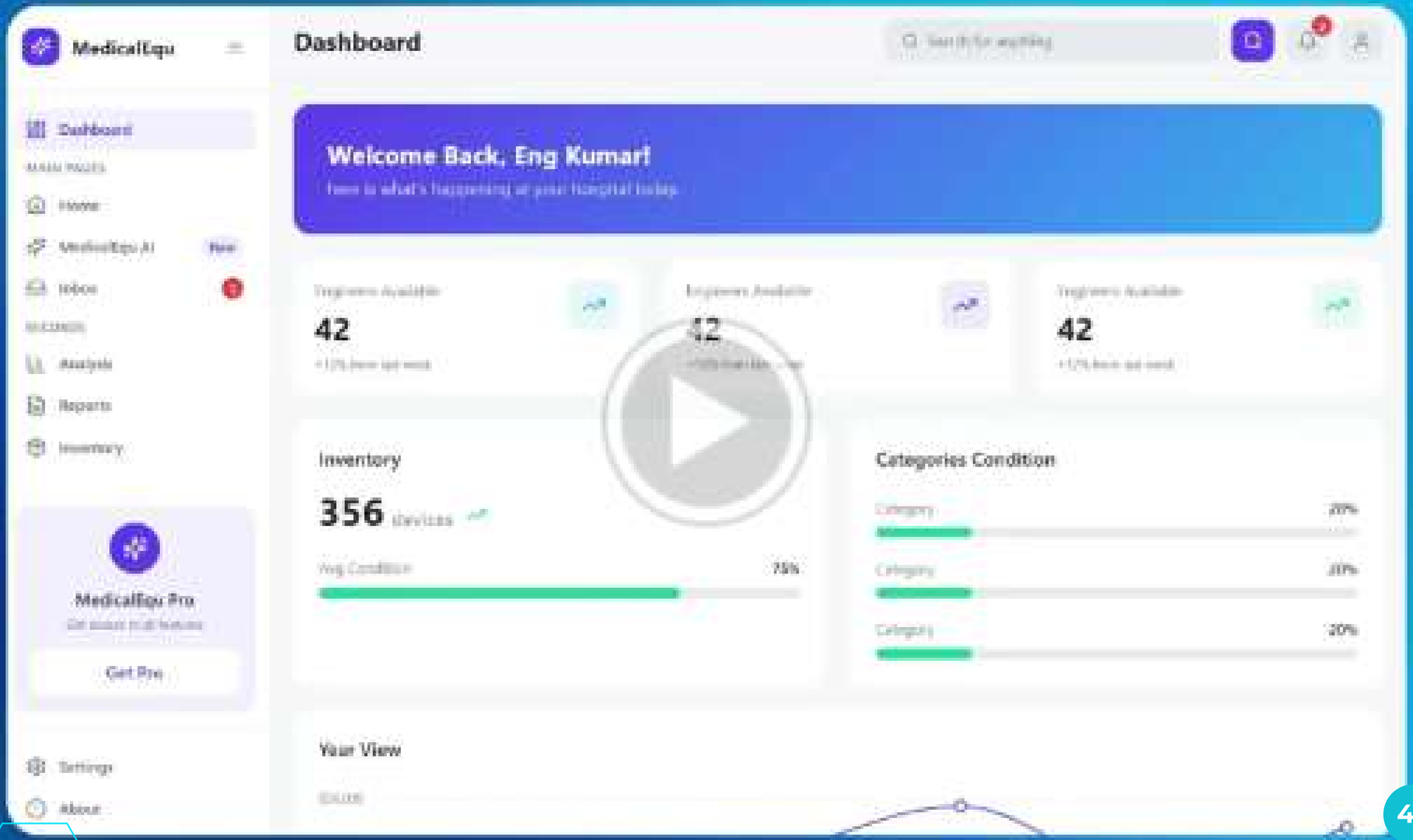
6. Secure System Architecture

- **Tools**: Spring Boot , Spring Security, JWT Auth, OAuth2
- Ensures secure authentication, data protection, and full audit logging.

7. Interactive User Interface

- **Tools:** React, Figma, Material UI
- Provides responsive and user friendly dashboards with real-time system updates.

Demo


MedicalEqu
Dashboard
Dashboard
Home
Inbox
Analysis
Reports
Inventory
MedicalEqu Pro
Get Pro
Settings
About
Welcome Back, Eng Kumar!
42
42
42
Inventory
356
Categories Condition
Your View

Conclusion

Conclusion

- The project presents an AI-powered medical equipment management system tailored for the Egyptian healthcare environment.
- Fills the gap in traditional medical device management
- The system enhances patient safety through predictive maintenance and risk

Conclusion

- It supports compliance with International Standards(ISO and GAHAR) , increasing hospital reputation
- Built to be affordable, scalable, and suitable for Egyptian hospitals.

Future Work

Completed Project Phases

July 2025 - November 2025

1.0

Requirements & Planning

Jul 27, 2025

2.1

Sep 21, 2025

Data Collection - Device Inventory

2.2

Data Collection - Maintenance History

Oct 5, 2025

2.3

Nov 2, 2025

Data Collection - Failure & Downtime Logs

2.4

Data Collection - Data Validation

Nov 16, 2025

Future Work

- Perform data cleaning and preprocessing to prepare high-quality datasets for AI training.
- Test the AI model thoroughly to ensure accuracy and reliability.
- Complete the development of the entire system, including frontend, backend, and AI integration.

Ongoing & Upcoming Project Phases

January 2026 - June 2026

3.0
Data Processing & Feature Selection
Jan 2026

Feb 2026
4.0
AI Maintenance System Development

5.0
Development Setup
Apr 2026

May 2026
6.0
Testing & Bug Fixing

7.0
Optimization & Security
Jun 2026

Jul 2026
8.0
System Integration

Future Work

- Train the AI model and evaluate its performance on the processed data.
- Integrate all components and test the full system workflow to ensure seamless operation.
- Conduct comprehensive system testing to identify and fix bugs.
- prepare deployment documentation.

References

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- React.js: react.dev

2. Backend & Security

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- Spring Security: spring.io/projects/spring-security

3. Database

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- PostgreSQL: postgresql.org/docs

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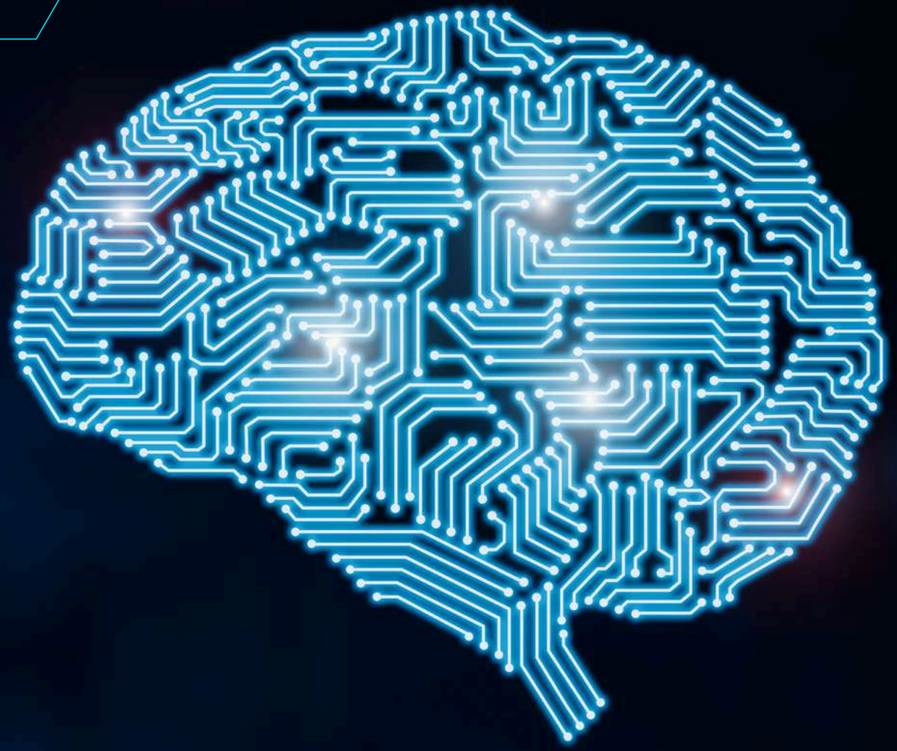

Thank You

Any Questions ?